

Q25 Peak voltage in secondary coil is 90 V.
Resistor R_2 and R_3 are in parallel connection,
 $\Rightarrow$ total resistance in circuit connected to the secondary coil is 120Ω .

$$\text{R.m.s. voltage in secondary coil, } V_{\text{r.m.s}} = \frac{90}{\sqrt{2}} \text{ V}$$

$$\text{R.m.s. current in secondary coil, } I_{\text{r.m.s}} = \frac{\frac{90}{\sqrt{2}}}{120} \text{ A}$$

$$\begin{aligned} \text{Average power dissipated in } R_1 \\ &= (I_{\text{r.m.s}})^2 \times 80 \\ &= 22.5 \\ &\approx 23 \text{ W} \end{aligned}$$

Answer: B

Q26 A and B seems to be the closest options. hf should be a form of energy rather than the